Alex Tang

925-596-5967 | yalext08@ucla.edu | linkedin.com/in/yalext | github.com/aelext

Education

University of California, Los Angeles (UCLA) — Los Angeles, CA
Bachelor of Science in Mathematics of Computation — *Expected Jun 2029*
- **Relevant Courses**: Introduction to Computer Science, Introduction to Discrete Structures

Amador Valley High School — Pleasanton, CA
Valedictorian (1/641) | GPA: 4.70 W / 3.98 UW — *Aug 2022 – Jun 2026*

Technical Skills

Computer Languages: Java, Python, C++, R, MATLAB
Tools: OR-Tools, Gurobi, ChromaDB, Ollama, Jupyter, MySQL

Work & Research Experience

English/Mathematics Tutor | *Kumon* — Jul 2026 – Sep 2026
- Tutored 50+ students in advanced mathematics curricula up to calculus

Complex Systems Research (SRA, UC Santa Barbara) | *MATLAB, Simulink* — Jun 2025 – Jul 2025
- Modeled physical effects of network delay and packet loss on 3-DOF robotic arm operational accuracy
- Co-authored a research paper detailing the degradation thresholds of network latency

Mathematical Modeling of Biological Systems (COSMOS, UC Davis) | *R* — Jul 2024
- Created ODE models to simulate real-world phenomena in epidemiology, ecology, and physiology
- Developed a mathematical model simulating biomass diffusion across channel-like landscapes

English/Mathematics Tutor | *Afficient Academy* — Apr 2024 – Jun 2024
- Tutored 20+ students in advanced mathematics curricula up to calculus

Technical Projects

BLE Wearable Step Counter (Bruin Software Engineers) | *C++, Arduino, Edge Impulse* — 2026 – Present
- Built a wearable step counter in Arduino C++ that streams live step counts to a phone over a custom BLE service
- Detected steps with accelerometer thresholds and a 300 ms debounce, gated by an on-device TinyML model

E-Sports Predictive Analytics Model (DataRes) | *Python, R* — 2026 – Present
- Conducting EDA on multiplayer match data to quantify impacts of spatial advantages and resources
- Training ML models evaluating how early-game momentum and player distributions predict final outcomes

Independent Project: Local RAG System | *Python, ChromaDB, Ollama, Jupyter* — 2026
- Built a local Retrieval-Augmented Generation (RAG) system with document embedding and querying
- Deployed open-source LLMs like qwen3 for local data processing without relying on external APIs

Independent Research Project: Anti-Poaching Model | *Python, OR-Tools, Gurobi, MySQL* — 2023 – 2026
- Applied linear and integer programming to route park rangers through Hell's Gate National Park hotspots
- Implemented a MySQL database to efficiently query and store mathematical model data constraints

Co-Author, NeuroPass: Advanced Precision Therapy for MS — 2025
- Designed an AI and ultrasound-guided quantum dot drug delivery concept for Multiple Sclerosis
- Earned a Toshiba ExploraVision Honorable Mention (top 10% of 15,000) as Co-Author

Leadership & Extracurriculars

Funding Lead | *Bruin Software Engineers (BSE)* — Nov 2026 – Present
- Research and apply for UCLA and SOLE grants and manage corporate sponsor relationships
- Maintain the sponsorship packet and member resume book distributed to industry partners

Co-President | *Music4America (M4A), Member 2019 – 2026* — 2024 – 2025
- Scaled non-profit operations, nearly doubling membership from 127 to 244 members
- Coordinated and performed in 30+ musical recitals at local senior centers and VA hospitals
- Directed community events and math camps, raising over $6,000 for local school music programs

Competitive Programmer | *Codeforces & USACO* — 2024 – Present
- Achieved Codeforces rating of 1726 (Expert tier); promoted to USACO Gold division (2025)
- American Invitational Math Exam (AIME) Qualifier: 121.5 on AMC 10A (Top 5%)